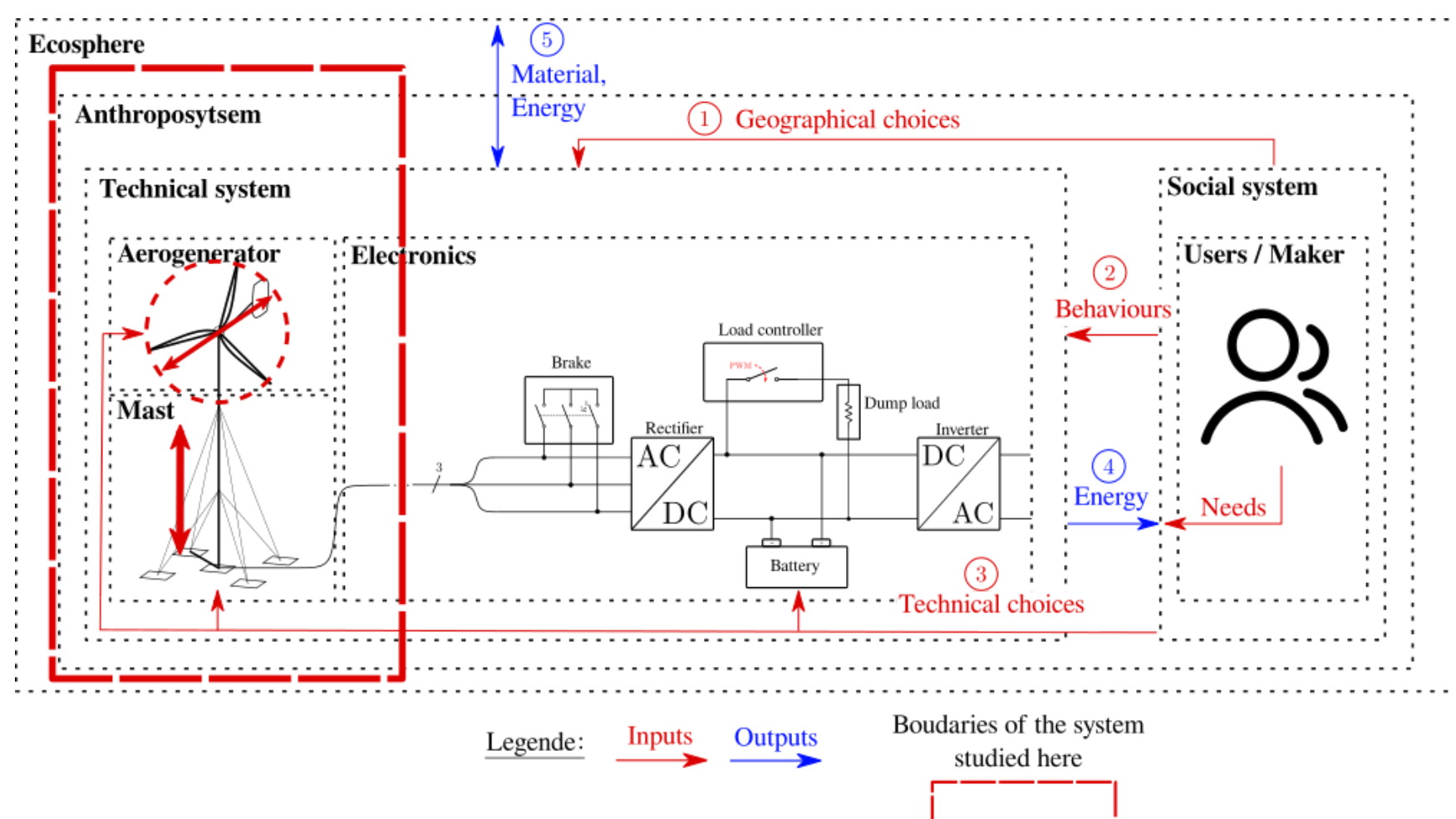


1. Introduction

This work evaluates the environmental performance of small wind turbines (SWTs) based on mast height and rotor diameter. Using parametric LCA and optimization, we explore design trade-offs to support sustainable, locally manufactured systems.

2. Methodology

Figure 1 : Framework of the study.



The framework used here is adapted from [1]. We use a narrower one (red rectangle on Figure 1), focused on the electro-mechanical part i.e. the aero generator and the structure. The levers under studied are : the diameter and the height of the wind turbine, the localisation and a production target.

This LCA is cradle-to-gate. Data was collected on end-of-life care, but it was not incorporated due to lack of time. However, this data enabled us to set the wind turbine's lifespan at 20 years.

FU : Total electricity production in kWh at the alternator output over 20 years on the European continent.

This LCA has been done on Brightway 2 with the database Ecoinvent 3.8 cut-off. We use the Ecological Footprint method for the single score.

3. Tools

Several tools have been used to calculate the LCA. Most of them are coded in Python. The Figure 2 sum up the interactions between them.

To calculate a continuous parametric inventory, we compute polynomial regressions on the existing designs in the handbook. For the generator, we create a 3D model of each design with the software OpenAFPM before running the regression. Then, the library lca.algebraic has been used to structure the parametric inventory.

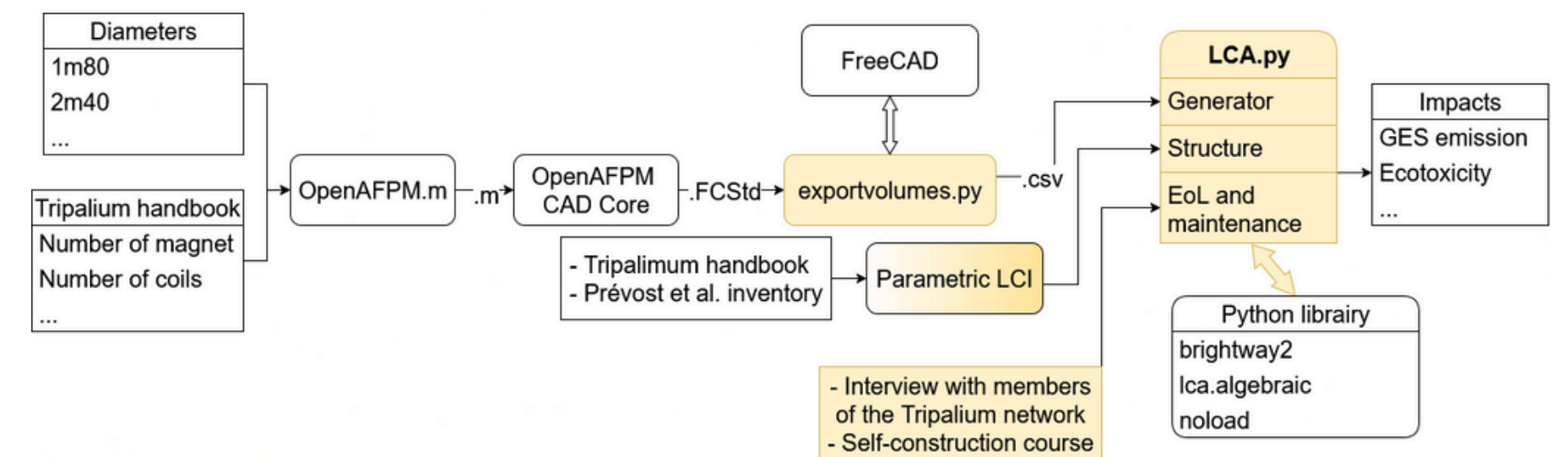


Figure 2 : Global structure of the interaction between tools. Yellow = tools coded

4. Results

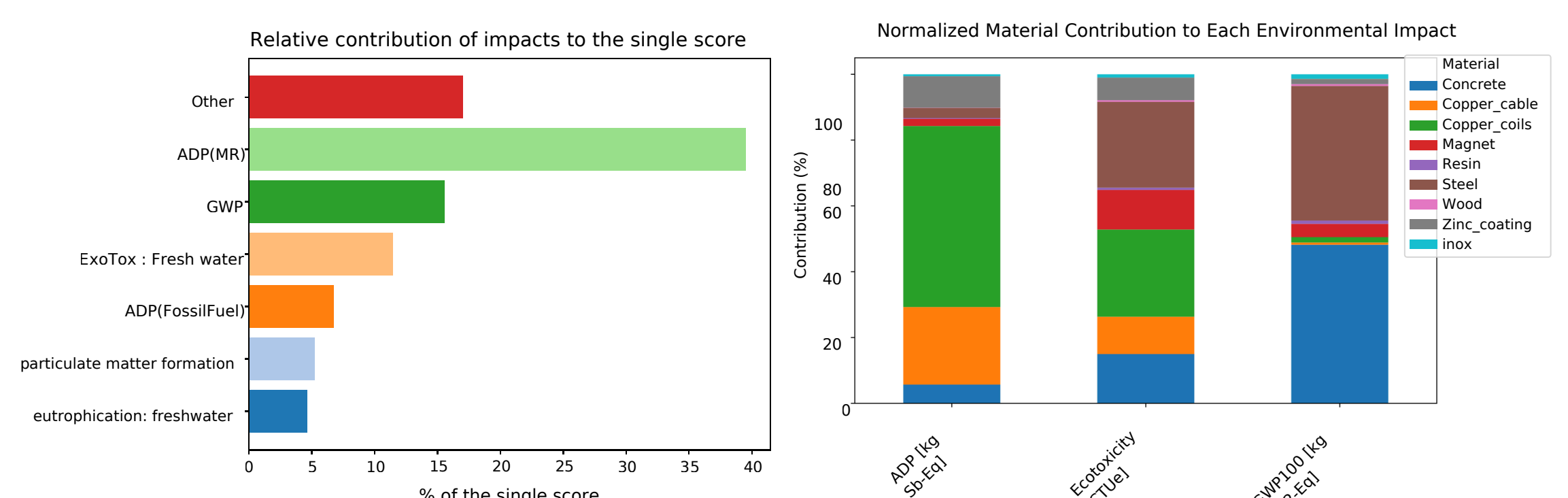


Figure 3 : Analysis of the impacts and materials contribution to the single score

We saw in Figure 3 that the main contributor to the single score is the use of minerals (ADP(MR)) which mainly due to the presence of copper. Next comes global warming, fuelled by steel and concrete. We can conclude that increasing the diameter of the wind turbine will mainly impact the ADP(MR), whereas the size of the mast will affect the GWP. There is therefore a trade-off between height and diameter.

To study this in more detail, we carried out a series of optimisations using the NoLoad library. For six different locations, i.e. different v_{moy} and z_0 values, we sought the configuration that minimised the single score per energy produced, with a production target ranging from 500 to 5,000 kWh per year.

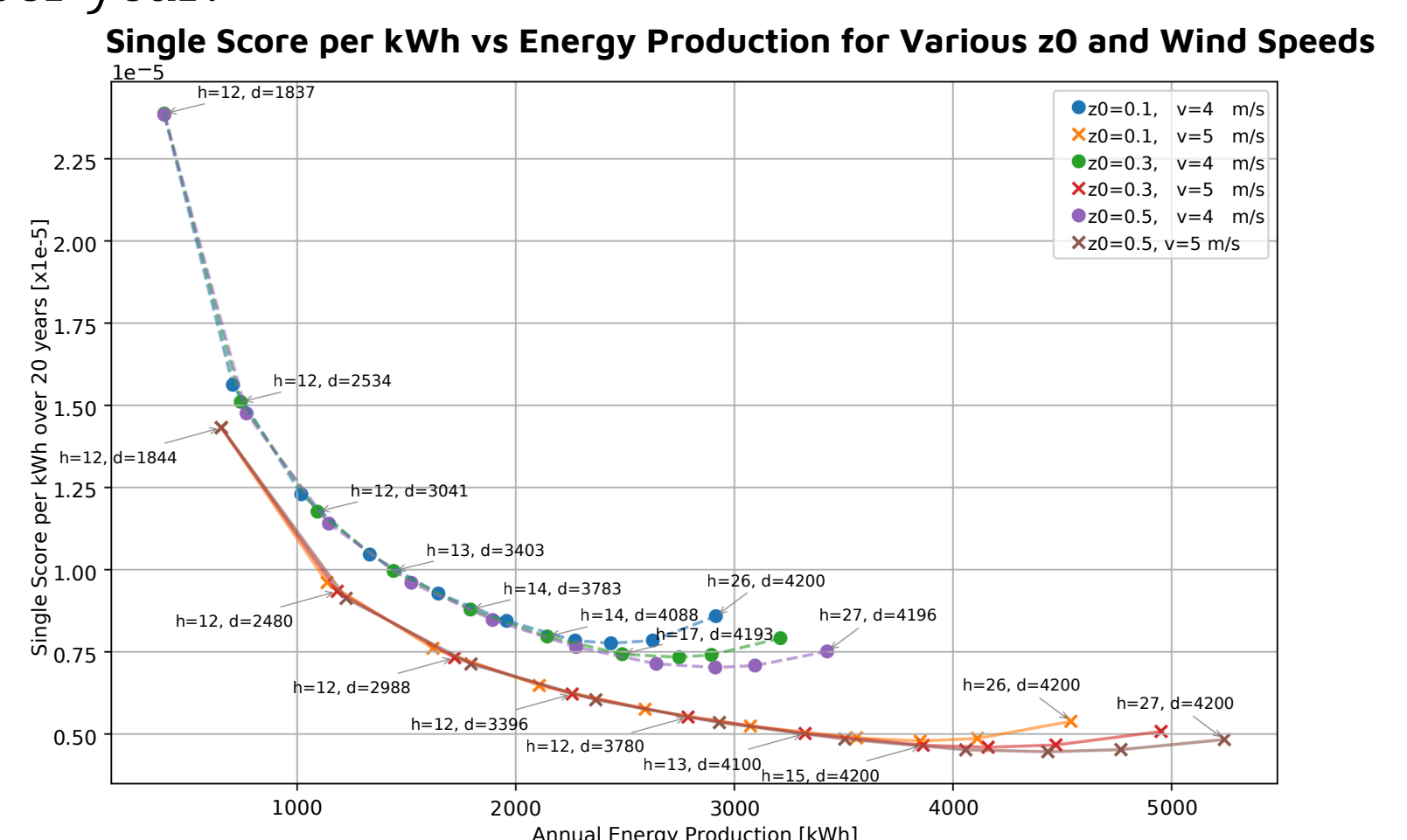


Figure 4 : Results of the optimisations.

The Figure 4 shows that the environmental efficiency decreases with the production. But the diameter reaches it upper bound in first. Then the extra production is only achieving by increasing the size of the mast, which leads to increasing the absolute impact quickly. Therefore, a large wind turbine on a small mast seems more interesting on an environmental point of view rather than a small one on a high mast.

[1] A. Prévost, F. Demichel, V. Léchappé, H. Helbling, R. Delpoux, and X. Brun, "A framework for technico-environmental optimization of small wind turbines," Applied Energy, vol. 377, p. 124502, Jan. 2025.